Scene Graph Generation Literature Review

Object Retrieval in Dynamic Workspace Environments

Purpose: Review SGG methods relevant to RGB-based semantic workspace understanding, identify useful architectural ideas, and define a model-independent direction for efficient SGG with selective semantic reasoning. | Important: No final SGG/VLM model is fixed. The reviewed methods are candidates and sources of design ideas.

1. SGG LITERATURE SUMMARY

Paper / Method	Core Contribution	Datasets / Benchmarks	Reported Evidence	Limitations	Relevance
REACT	YOLO-based Decoupled Two-Stage SGG using DTS, DCS and efficient predicate prototype learning to reduce unnecessary relationship computation.	PSG; YOLOv8/v9/v10 configurations	YOLOv8m: 23.9 F1@K, 53.1 mAP50, 32.5 ms, 43.3M parameters; reported ~2.7× speedup and ~5.5× fewer parameters than compared SOTA.	Static; fixed predicate vocabulary; hardware/dataset dependent; no temporal memory.	Strong Stage-1 efficiency baseline; DTS + DCS are highly relevant.
REACT++	Extends REACT with DAMP, AIFI, CARPE, Geometry RoPE and DCS for efficient local, global, semantic and spatial relation reasoning.	PSG and reported VG-related comparisons	35.8M parameters; 25.9 ms without DCS; 19.4 ms with DCS; ~20% mean-Recall improvement; DCS gives ~66.5% average latency reduction with ~1% relative F1@K loss.	Static SGG; dataset bias; sensitivity to noisy representations; hardware-specific latency.	Current strongest Stage-1 candidate based on literature, not final.
FCSGG	Fully convolutional SGG using center-based object representation and Relation Affinity Fields (RAFs) for spatial relationship modelling.	Visual Genome	ResNet50-4S-FPN×2: ~36M parameters, ~40 ms/image at 512×512 on GTX 1080 Ti; competitive Recall/mR/zero-shot Recall.	Closed vocabulary; semantic ambiguity; no temporal memory; pair reasoning remains challenging.	Important efficiency and spatial-relation baseline; RAF is a useful idea.
Fast Contextual SGG	Uses object categories, bounding boxes and relative geometry to predict plausible relations cheaply, followed by selective visual verification.	Visual Genome; PSG; OpenImages-V6	YOLOv5l + C-SGG: 33.5 FPS; YOLOv5l + CV-SGG: 6.4 FPS; Rmask retained the true predicate in top-10 in ~98% of reported VG tests.	Context priors are dataset-dependent; context alone misses fine visual interactions; the contextual mask may suppress a needed predicate if C-SGG fails to rank it among the selected candidates.	Very relevant candidate-filtering idea before expensive reasoning.
From Pixels to Graphs	Uses BLIP/InstructBLIP-based image-to-sequence generation for open-vocabulary SGG beyond a fixed predicate vocabulary.	PSG; OpenImages-V6; Visual Genome	Evaluated for novel predicates and open-vocabulary graph generation against conventional SGG/VLM approaches.	VLM inference is expensive; low resolution can hurt small-object localization; grounding/hallucination concerns; static.	Strong Stage-2 VLM semantic reasoning reference.
Open3DSG	Creates an open-vocabulary, queryable 3D scene graph by aligning 2D VLM knowledge with 3D representations.	3DSSG / ScanNet	2D–3D ablation: R@5 object 0.57, predicate 0.45; R@3 object 0.63, predicate 0.37.	Requires 3D point clouds/RGB-D; higher computation; outside current RGB-only target.	Evidence for queryable open-set relationship reasoning, not a direct deployment candidate.

1. A SURVEY FOR SCENE GRAPH GENERATION BASED ON PRE-TRAINED MODELS

Overview

This survey studies how pre-trained Vision-Language Models (VLMs) and Large Language Models (LLMs) are being incorporated into Scene Graph Generation to improve semantic understanding, open-vocabulary reasoning and long-tail relationship prediction.

Main Approaches / Components

1. Knowledge Prior & Feature Enhancement

- VLMs such as CLIP provide visual-language semantic features.
- LLMs can provide semantic knowledge or relationship priors.
- These features are combined with conventional SGG models to improve difficult or rare relationships.
- Examples include approaches such as CaCao, VLPrompt and LED.

2. LLM as Relationship Reasoner

- Objects are first identified from the image.
- Their visual/semantic representations are given to an LLM.
- The LLM reasons about possible relationships between objects.
- This enables more flexible and open-vocabulary relationship prediction.

3. End-to-End Multimodal SGG

- An MLLM receives the image and an instruction/query.
- It directly generates the scene graph or structured representation.
- This removes the need for separate traditional SGG modules but increases computational cost and the risk of hallucinated or incorrectly grounded relationships.

Input → Output

```
Image → Objects / Visual Features → VLM / LLM Reasoning → Semantic Relationships → Scene Graph
```

Relevance to Our Project

The most important takeaway is that VLMs do not necessarily need to replace the complete SGG system. They can act as a semantic reasoning layer when conventional SGG cannot confidently determine a relationship.

2. REACT — REAL-TIME EFFICIENCY AND ACCURACY COMPROMISE FOR TRADEOFFS IN SGG

Overview

REACT focuses on making Scene Graph Generation faster and more computationally efficient while maintaining relationship prediction performance.

Main Components

1. YOLO — Object Detection

- Detects objects in the image.
- Provides object categories, bounding boxes, confidence scores and feature information.

2. DTS — Decoupled Two-Stage

- Separates the system into: Stage 1: Object Detection, Stage 2: Relationship Prediction.
- This allows relationship reasoning to operate directly on detected objects.

3. DCS — Dynamic Candidate Selection

- Detected objects are ranked according to their confidence scores.
- The highest-confidence candidates are selected.
- Directed subject-object pairs are then formed among these selected objects.
- This reduces the number of candidate relationships processed.

4. GloVe

- Predicate words such as on, behind, holding, etc. are represented using GloVe word embeddings.
- This provides semantic information about the predicate vocabulary.

5. Efficient Prototype Learning

- Each predefined predicate has a learned prototype/reference representation.
- Visual features of a subject-object pair are compared against these predicate prototypes.
- The model produces scores for possible predicates.

Input → Output

```
Image → YOLO → Detected Objects + Bounding Boxes → DCS → Selected Objects → Directed Object Pairs → Pair Features + Predicate Prototypes → Relationship Scores → <Subject, Predicate, Object> → Scene Graph
```

Relevance to Our Project

REACT is an important efficient SGG baseline. Its DTS + DCS + prototype-learning approach is particularly relevant to our requirement for computationally efficient semantic relationship prediction.

3. REACT++ — EFFICIENT CROSS-ATTENTION FOR REAL-TIME SGG

Overview

REACT++ improves REACT by reducing feature-extraction cost and strengthening semantic, contextual and spatial relationship reasoning.

Main Components

1. YOLO + NMS + DCS

- YOLO detects objects and provides feature maps.
- NMS removes duplicate detections.
- DCS selects useful high-confidence objects before relationship reasoning.

2. DAMP — Detection-Aware Multi-scale Pooling

- Traditional RoI Align extracts features from every detected bounding box, which is computationally expensive.
- DAMP instead uses the known detection locations to directly gather features from YOLO's existing multi-scale feature maps.
- It uses YOLO feature levels such as P3, P4 and P5.
- Therefore, the model avoids repeatedly performing expensive RoI feature extraction.

3. AIFI — Attention-based Intra-scale Feature Interaction

- DAMP mainly provides object-specific/local information.
- AIFI — enables global/contextual feature interaction through attention.
- This provides global scene context, helping the model understand relationships using information beyond the individual object pair.

4. CARPE — Cross-Attention Rotary Prototype Embedding

- Uses cross-attention between object-pair representations and predicate prototypes.
- The model learns which predicate representations are relevant to the particular subject-object pair.
- It treats subject and object representations separately, helping model the direction of relationships.

5. Geometry RoPE

- Incorporates bounding-box geometry into the attention mechanism.
- Provides information about the relative position and spatial arrangement of the subject and object.
- Therefore, relationship prediction uses both semantic information and geometry.

Input → Output

```
Image → YOLO → Objects + Bounding Boxes + Feature Maps → NMS + DCS → DAMP (Object Features) + AIFI (Global Context) → CARPE (Cross-Attention + Predicate Prototypes + Geometry RoPE) → Relationship Scores → <Subject, Predicate, Object> → Scene Graph
```

Relevance to Our Project

REACT++ is currently our strongest Stage-1 SGG candidate because it combines object detection, efficient feature extraction, candidate reduction, semantic predicate modelling and spatial reasoning. However, its reported latency must be experimentally verified on our target hardware.

4. FULLY CONVOLUTIONAL SCENE GRAPH GENERATION — FCSGG

Overview

FCSGG proposes a fully convolutional, one-stage approach where objects and relationships are predicted directly from convolutional feature maps.

Main Components

1. CNN Backbone

- Uses backbones such as ResNet-50/HRNet to extract visual features.

2. FPN

- Combines features from multiple resolutions.
- Helps represent objects of different sizes.

3. Center-based Object Representation

- Instead of relying primarily on region proposals, objects are represented using their center points.
- The network predicts object centers, categories and size information.

4. RAF — Relation Affinity Field

- The main contribution of FCSGG.
- A RAF represents the relationship between two objects as a 2D vector field connecting their centers.
- The vector provides directional/spatial information from the subject toward the object.
- RAF primarily represents the spatial/directional relationship between object centers; the resulting representation is then used for predicate prediction.

5. Path Integral

- Once two objects are identified, the model examines the RAF along the path between their centers.
- The accumulated information is used to predict the relationship between them.

Input → Output

```
Image → CNN + FPN → Object Centers + Object Classes + Relation Affinity Fields → Path Integral → Predicate Prediction → <Subject, Predicate, Object> → Scene Graph
```

Relevance to Our Project

FCSGG is an important efficiency and spatial-representation baseline. Its RAF concept is particularly interesting because it represents spatial relationships without requiring expensive feature extraction for every object pair. However, it is less modular than REACT/REACT++ and still relies on a predefined relationship vocabulary.

5. FAST CONTEXTUAL SCENE GRAPH GENERATION WITH UNBIASED CONTEXT AUGMENTATION

Overview

This paper introduces a context-first approach where relationships are initially predicted using object categories and spatial information, without visual features. Visual reasoning is selectively applied to the most likely predicates identified by the contextual model.

Main Components

1. Object Detector

- An external detector such as YOLOv5 provides object category and bounding box.
- The detector is not the main contribution of the paper.

2. Context Description

- For every subject-object pair, the model combines subject category, object category, subject bounding box, object bounding box, and relative spatial information.
- Object categories are represented using GloVe embeddings.
- Importantly, this stage uses no visual features.

3. CKN — Context Knowledge Network

- Takes the context representation as input.
- Uses fully connected layers to predict probabilities for possible predicates (e.g., Person + Cup + Spatial Arrangement → holding: high, near: high, wearing: low).

4. Context Augmentation

- Bounding boxes are slightly modified during training while preserving the relationship.
- This creates additional contextual variations and helps with rare/long-tail predicates.

5. Rmask

- Selects the most promising contextual predicates.
- Only these candidates are passed to the visual verification stage.

6. VDN — Vision Differentiation Network

- Uses visual information from the image and subject/object regions.
- Verifies whether the contextually predicted relationship is actually supported by the image.

Input → Output

```
Image → YOLOv5 → Object Categories + Bounding Boxes → Context Description → CKN / C-SGG → Likely Predicates
→ Rmask → VDN / CV-SGG → Visual Verification → Final Predicate → Scene Graph
```

Relevance to Our Project

This paper provides an important architectural principle: Use cheap contextual reasoning first and expensive visual reasoning only when necessary. This is highly relevant to our idea of keeping the main SGG lightweight and invoking a more expensive VLM only when the existing scene graph cannot resolve the requested relationship.

6. FROM PIXELS TO GRAPHS: OPEN-VOCABULARY SCENE GRAPH GENERATION WITH VISION-LANGUAGE MODELS

Overview

This paper explores using Vision-Language Models to generate scene graphs with open-vocabulary relationships, rather than restricting relationship prediction to a fixed predicate list.

Main Components

1. BLIP / InstructBLIP

- Pre-trained VLMs connect image information with language.
- They can generate textual descriptions of objects and relationships.

2. PGSG

- Uses an image-to-sequence generation approach for SGG.
- Instead of selecting only from predefined predicate classes, relationships can be expressed through generated language.

3. Entity Grounding

- Generated object descriptions must be connected to actual objects in the image.
- This links the generated language representation with the visual entities.

4. Category Conversion

- Converts the grounded language output into a structured scene graph.

Input → Output

```
Image → BLIP / InstructBLIP → Generated Objects + Relationships → Entity Grounding → Structured Graph Conversion → Open-Vocabulary Scene Graph
```

Relevance to Our Project

This is more relevant to our Stage-2 VLM layer than to Stage 1. It demonstrates how a VLM can handle relationships that may not exist in the fixed SGG predicate vocabulary. However, the computational cost and reliability of VLM-based generation make it unsuitable as the continuously running perception module for our edge-oriented system.

7. OPEN3DSG — OPEN-VOCABULARY 3D SCENE GRAPHS

Overview

Open3DSG creates a queryable open-vocabulary 3D scene graph by combining 3D information with vision-language knowledge.

Main Components

1. PointNet / 3D GNN

- Extracts and propagates features from 3D point-cloud data.
- Provides geometric and spatial information.

2. OpenSeg

- Provides open-vocabulary visual features from 2D images.

3. BLIP / InstructBLIP

- Provides pretrained vision-language knowledge.

4. 2D–3D Alignment / Knowledge Distillation

- Transfers semantic knowledge from 2D vision-language representations into the 3D representation.

5. CLIP

- Enables semantic/open-vocabulary object querying.

6. Q-Former + LLM Decoder

- Connects the scene representation to language-based querying and reasoning.

Input → Output

```
RGB-D / Point Cloud → 2D Visual Features + 3D Features → 2D–3D Alignment → Language-Aligned 3D Scene Graph →
Natural Language Query → Open-Set Object / Relationship Result
```

Relevance to Our Project

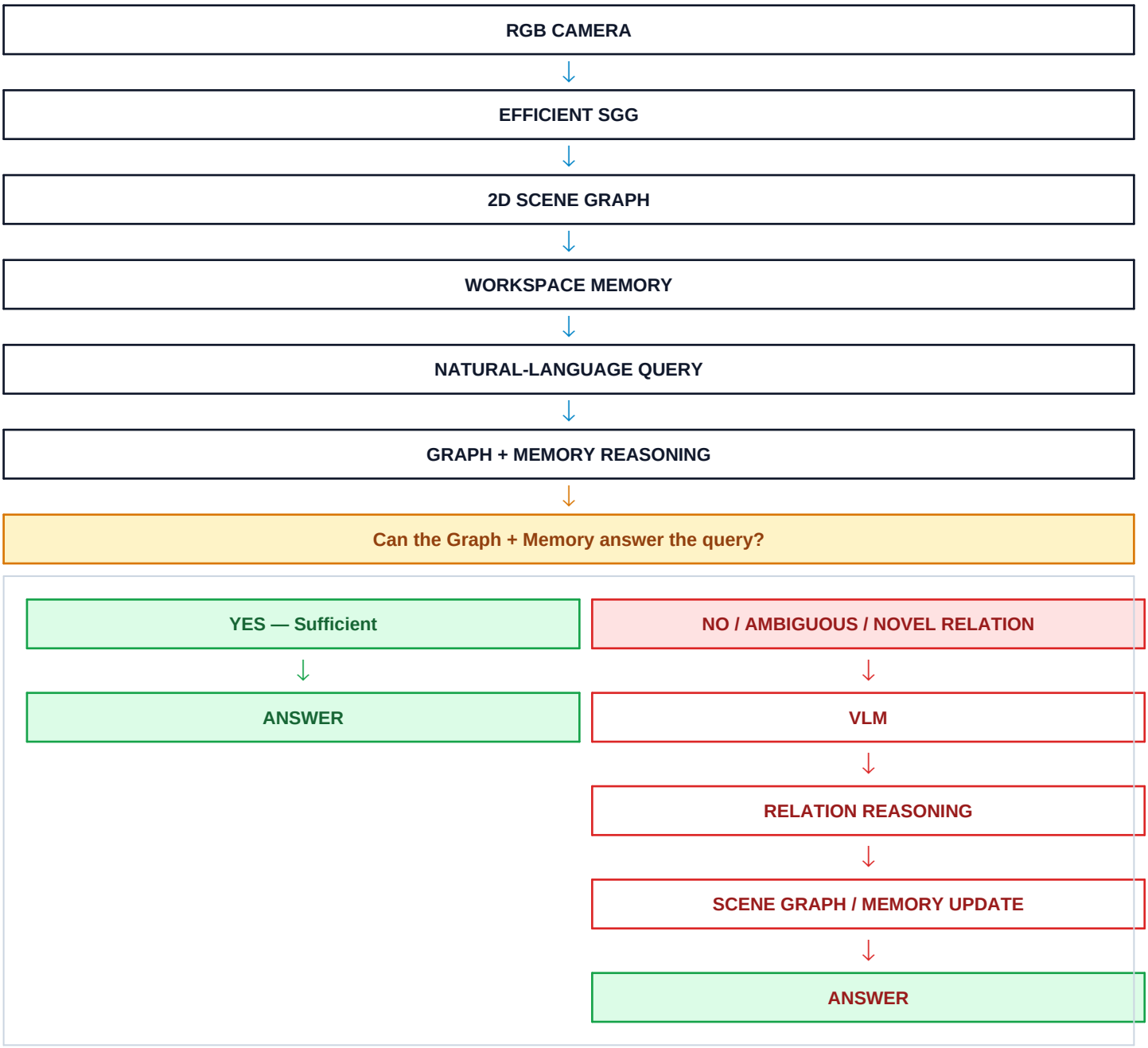
Open3DSG is not directly suitable for our current RGB-only target because it depends on 3D/RGB-D information. Its most useful contribution for us is the concept of a queryable scene graph capable of handling open-set relationships.

8. PROPOSED PIPELINE

Based on the individual paper review, the proposed pipeline is deliberately **model-agnostic**. The report shows the conceptual modules rather than locking the design to REACT++ at this stage.

> *Efficient SGG handles normal scene understanding continuously. The VLM is only invoked when the existing Scene Graph + Memory cannot confidently resolve the user's query.*

Pipeline Overview



1. RGB Camera

Purpose

The camera continuously observes the workspace/room. The system initially uses RGB-only perception, avoiding the additional cost and complexity of depth sensors.

Input

```
Physical workspace scene.
```

Output

```
RGB image/frame.
```

Example

```
Workspace → RGB Frame
```

2. Efficient SGG

Purpose

Convert the RGB image into a structured understanding of the scene. The selected SGG model will handle the required object detection + relationship prediction internally. We are currently considering models such as REACT++ and FCSGG, but the final model is not decided.

Input

```
RGB image.
```

Output

```
Objects + bounding boxes + semantic relationships.
```

Example

```
Laptop, Notebook, Phone, Table
Laptop → beside → Notebook
Phone → near → Laptop
Notebook → on → Table
```

3. 2D Scene Graph

Purpose

Represent the detected scene in a structured form. The graph consists of nodes (objects) and edges (relationships). Each relationship can be represented as <Subject, Predicate, Object>.

Input

```
Objects + predicted relationships from SGG.
```

Output

```
Structured 2D Scene Graph.
```

Example

```
<Laptop, beside, Notebook>
<Notebook, on, Table>
<Phone, near, Laptop>
```

4. Workspace Memory

Purpose

Maintain the scene information over time instead of treating every frame independently. It can potentially store object identity, relationships, previous observations and last known location. This allows future queries such as “Where were my earphones

last seen?” This is a proposed component; its exact implementation is not finalized.

Input

```
Current Scene Graph + previous memory.
```

Output

```
Updated Workspace Memory.
```

Example

```
t1: Earphones → beside → Notebook
t2: Earphones → near → Laptop
```

5. Natural-Language Query

Purpose

Allow the user to interact with the system naturally, e.g. “Where is my charger?”, “What is beside the laptop?”, “Find the black object near the notebook.”, “Where was my phone last seen?”

Input

```
Natural-language query.
```

Output

```
Query constraints / intended relationship.
```

Example

```
"What is beside the laptop?"
Reference object = Laptop
Relationship = beside
```

6. Graph + Memory Reasoning

Purpose

First try to answer the query using the information that is already available. This is the important computational-saving decision point.

Input

```
Natural-language query + Scene Graph + Workspace Memory.
```

Output

```
Answer found, or: Insufficient / ambiguous / unknown relationship.
```

Example

```
Query: "What is beside the laptop?"
Scene Graph: Laptop → beside → Phone
System answers directly: Phone
```

7. VLM — Selective Reasoning

Purpose

The VLM is not intended to run continuously. It is activated only when the graph cannot answer the query, the relationship is ambiguous, the required relationship is outside the SGG predicate vocabulary, or higher-level semantic reasoning is required.

Input

```
RGB image + Query + Relevant scene information.
```

Output

```
Refined/previously unknown relationship.
```

Example

```
Query: "Which object is supporting the phone?"
Scene Graph: Phone → near → Laptop; Phone → on → Table
Scene Graph → VLM → Visual + language reasoning → <Phone, supported-by, Table>
```

8. Scene Graph / Memory Update

Purpose

If the VLM discovers a useful relationship, it should not necessarily disappear after answering the query. The new information can be added to the representation, since the system can potentially use the newly learned relationship for future queries.

Input

```
VLM-derived relationship.
```

Output

```
Updated Scene Graph / Workspace Memory.
```

Example

```
Existing Graph: Phone → near → Laptop
VLM discovers: Phone → supported-by → Table
→ Updated Graph
```

9. Final Object Retrieval

Purpose

The final stage converts the semantic answer into an object/location that the user or future robot can use. Future robotics integration could then use this output for object retrieval, robot planning and manipulation, but robotics is outside the current core pipeline.

Input

```
Identified object + bounding box / spatial information.
```

Output

```
Object identity + location.
```

Example

```
"Where is the charger?" → Scene Graph + Memory → Charger → Bounding Box / Image Coordinates → Object Location
```

One-Page Summary

Stage	Purpose	Input	Output
RGB Camera	Observe workspace	Physical scene	RGB frame
Efficient SGG	Detect objects and predict relationships	RGB frame	Objects + relationships
2D Scene Graph	Structured scene representation	Objects + relationships	Object-relation graph
Workspace Memory	Maintain information over time	Current graph + previous memory	Updated memory
Natural-Language Query	Understand user's request	User query	Query constraints
Graph + Memory Reasoning	Try to answer without expensive reasoning	Query + graph + memory	Answer / unresolved query
VLM	Resolve ambiguous/unknown semantic relationships	Image + query + relevant context	Refined relationship
Graph/Memory Update	Store newly discovered information	VLM result	Updated graph/memory
Object Retrieval	Locate requested object	Identified object + spatial data	Object location

The Core Concept

> *The system first relies on an efficient SGG and workspace memory for normal queries. Only when the existing structured representation is insufficient does it invoke a computationally expensive VLM to perform higher-level relationship reasoning and update the memory.*

This is the most important sentence on the pipeline page, because it explains why we are combining SGG + VLM instead of simply using a VLM for everything.